

Dipti Ghiya

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SUMMARY

Backend and systems software engineer with 6 years of experience building microservice infrastructure and event-driven systems. Designed and shipped Kafka-based pipelines, proactive monitoring services, and agentic AI workflows. Pursuing an M.S. in Computer Engineering at SJSU, deepening expertise in operating systems, distributed systems, and computer architecture.

SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, C, SQL
- **Backend & Systems:** Microservices, REST APIs, Apache Kafka, distributed systems, concurrency, multithreading, memory management, OS internals
- **Cloud & DevOps:** AWS (EC2, S3), Kubernetes, Docker, Jenkins, CI/CD, Git, Linux, Prometheus
- **Frameworks & Tools:** Spring Boot, Hibernate, React, Angular, Node.js, FastAPI, LangChain
- **AI & Agents:** Multi-agent systems, LLM orchestration, RAG, prompt engineering, Gemini API, Groq API

EXPERIENCE

Avalara (Senior Software Engineer), Pune, India

03/22 - 07/25

- Designed and shipped a proactive monitoring service across 7+ regions; automated external API health checks, suspended interactions during outages, and eliminated ~30 min/incident of manual debugging by surfacing dependency status on the portal.
- Built and maintained Java, Python, and Kafka microservices handling 5M+ e-invoicing events/year; improved reliability through asynchronous processing and resilient error handling.
- Built 3+ agentic AI workflows using Claude to extract planned downtime windows and detect risky API contract changes before release; demoed to CTO, reducing manual release review effort.
- Resolved 100+ P0 and critical security vulnerabilities (SQL injection, authorization flaws) across core services over 6 months during an acquisition audit, meeting compliance requirements within agreed SLAs.
- Served as Scrum Master for a 6–8 member backend team; drove on-time story delivery from ~65% to 85%+ across multiple sprints.

Accenture (Software Engineer), Pune, India

09/19 - 03/22

- Built Java, Spring Boot, Hibernate, and SQL microservices for term deposit and credit card workflows, supporting daily transaction processing for a major European bank.
- Optimized complex SQL queries and data access patterns, reducing slow-query execution times and ensuring compliant handling of sensitive customer data.
- Designed and implemented a UFT automated testing framework covering core customer journeys end-to-end, increasing regression coverage and reducing production defects per release.

EDUCATION

San Jose State University

San Jose, CA

M.S. Computer Engineering

Coursework: System Software, Advanced Computer Design, Computer Architecture, Computer Networks

Savitribai Phule Pune University

Pune, India

B.E. Electronics & Telecommunications

PROJECTS

- **Multi-Agent Security Vulnerability Scanner** | *Python, Groq, Gemini, Semgrep, FastAPI*: Built a 3-agent pipeline (recon, vulnerability analysis, report generation) to autonomously scan GitHub repos; combined Semgrep static analysis with LLM reasoning to detect attack chains and generate structured remediation reports. Implemented dual LLM client with round-robin load balancing and automatic fallback on rate limiting.
- **16-bit CPU ISA, Emulator & Assembler** | *C, Systems Programming*: Designed a custom 16-bit ISA with CALL/RET instructions and SP-based stack; built a C-based emulator modeling ALU, control unit, memory, and MMIO with a two-pass assembler; validated with Recursive Factorial and Timer programs.
- **TCP over Long-RTT Links in ns-3** | *ns-3, C++, Networked Systems*: Simulated TCP behavior across local, long-distance, and satellite-like links by varying RTT at fixed link speed; demonstrated how bandwidth-delay product limits usable throughput under standard TCP congestion control.

HONORS AND AWARDS

- **Avalara – Recognition Award**: Recognized for Scrum leadership, sprint accuracy, and mentoring new team members; supported hackathon presentations and maintained strong JIRA hygiene across the team.
- **Accenture – Apex Award (Extra Miler) & Project Level Award**: Awarded for automation testing contributions, on-time client delivery, and stepping up during periods with limited QA support.